

must mean that the value of the fixed side must also be par at the inception of the trade. That is, the swap rate is *the* fixed rate that makes the value of the fixed rate bond par when using the swap curve to discount cash flows.

The implications of this realization are extremely important. This means that the mid-market swap rates on a composite page like the one shown in Table 1.1 define a sequence of bonds that are each worth par, and the swap curve is therefore referred to as being a *par yield curve*.¹³

One nice aspect of par yield curves is that they lend themselves well to bootstrapping. This technique can be used in order to determine *zero coupon rates* and *forward rates* for any maturity that are implicit in the curve:

- *Zero coupon rates* are used to discount a single cash flow received at some point in time in the future back to today. Assuming annual compounding, if $zc(t)$ is the zero coupon rate for cash flow received at time t , then the present value of \$1 received at time t is

$$\frac{\$1}{[1 + zc(t)]^t}.$$

- Forward rates of interest are rates that can be locked in for future time periods, as determined by zero coupon rates. In the case of a swap, the forward rates of interest calculated are LIBOR forwards. It is common in the swaps market for people to think of forward rates as expressing the market's expectation of where LIBOR will be in the future, and we will employ this interpretation as well.¹⁴

Once again assuming annual compounding, we will denote by $f(t, t+1)$ the forward LIBOR rate between times t and $t+1$. When we say that a forward rate is a rate of interest that can be locked in for future periods, this means that \$1 to be received at time $t+1$ is worth the same as

$$\frac{\$1}{1 + f(t, t+1)}$$

at time t .

We will use zero coupon and forward rates to value every swap that we encounter from here on out.

¹³If we really wanted to be a stickler, we could say that a swap transacted on either the bid side or the offer side is not worth zero, since the dealer has presumably made some money on the transaction. So to avoid this criticism we'll just talk about a swap with a mid-market swap rate as having zero value.

¹⁴Empirical evidence may refute the validity of the assumption that a forward rate is equal to the expectation of the future spot rate. Fama (1984a, 1984b) wrote two early works examining this question in the interest rate markets. Despite raising doubts about the appropriateness of forward rates as expected future spots rates, most market participants fully adopt this usage, and we will too.

3.2.1 A Stylized Example

We are going to work with a particular stylized example in which we assume for simplicity that we live in a world in which the standard convention in swaps is that the fixed side pays annually 30/360, and the floating side is 1-year LIBOR, which pays annual 30/360 as well. We will also assume that yields on Treasuries are quoted annual bond and that swap spreads are quoted on the same basis. These assumptions are only made for simplicity; we could definitely assume that the standard semi bond fixed side and quarterly money floating side conventions apply, but that would just give us more numbers to calculate for a minimal pickup in modeling flexibility.

Suppose we walk in one morning and look on a composite page depicting our stylized example and observe the mid-market rates and spreads shown in Table 3.5. Each of the swap rates in the final column of Table 3.5 defines a fictitious bond that is worth par today. The cash flows of each of the four fictitious bonds (with an assumed principal of \$100) are depicted in Table 3.6. The present value of each of these bonds must be par, or \$100, today.

Now let's remember the big picture here: we are going to use these swap rates (and the par bonds they imply) to determine zero coupon rates and then forwards.

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	4.80%	20	5.00%
2	4.88%	23	5.11%
3	5.05%	25	5.30%
4	5.19%	28	5.47%

Table 3.5: Swap Curve Found on a Composite Page

Maturity	Swap Rate	1-Year Bond	2-Year Bond	3-Year Bond	4-Year Bond
1	5.00%	105.00	5.11	5.30	5.47
2	5.11%		105.11	5.30	5.47
3	5.30%			105.30	5.47
4	5.47%				105.47

Table 3.6: Par Bond Representation

Finding Zero Coupon Rates

To find the zero coupon rates, we proceed iteratively: we solve for one zero coupon rate at a time, then iterate to get the next one. Here's how it goes:

- The 1-Year Zero Coupon Rate is simply equal to the 1-Year Swap Rate:
 - The 1-year bond that pays \$105 in a year must be worth \$100 today. This implies that the 1-year zero coupon rate, $zc(1)$, must satisfy

$$\$100 = \frac{\$105}{1 + zc(1)}.$$

That is, the 1-year zero coupon rate, $zc(1)$, is 5%.¹⁵

- To find the 2-Year Zero Coupon Rate:
 - Working now with the 2-year bond that has a coupon of 5.11%, use the 1-year zero coupon rate to find the present value of the first-year coupon, which is $\$5.11/1.05 = \4.867 .
 - Since the 2-year bond is worth par today, and we know the present value of the first-year cash flow must be \$4.867, this implies the present value of the cash flow occurring in the second year must equal $\$100 - \$4.867 = \$95.133$.
 - Thus, the 2-year zero coupon rate must satisfy

$$\$95.133 = \frac{\$105.11}{[1 + zc(2)]^2},$$

or $zc(2) = 5.113\%$.

- To find the 3-Year Zero Coupon Rate:

¹⁵Note that in our (relatively simplistic) stylized example with annual 30/360 coupons on both sides, we could have concluded that the 1-year zero coupon rate was 5% in a different manner. Note that it must be the case that 1-year LIBOR today is 5%. By definition, the 1-year swap rate is the fixed rate that someone is willing to pay or receive versus LIBOR in a 1-year swap. That is, the 1-year swap rate is the fixed rate such that a 1-year swap is an NPV = 0 transaction. Since the 1-year swap rate is 5%, and since each side of the swap pays annual 30/360 coupons, we must conclude that 1-year LIBOR today is also 5%. Since 1-year LIBOR today is 5%, this means that \$1 invested today at LIBOR for a year grows to \$1.05 at the end of year. Equivalently, this means that the present value of \$1 to be received in a year is \$1/1.05, which is to say that the 1-year zero coupon rate is 5%.

- Working now with the 3-year bond that has a coupon of 5.30%, use the 1-year zero coupon rate to find the present value of the first-year coupon, which is $\$5.30/1.05 = \5.048 , and use the 2-year zero coupon rate to find the present value of the second-year coupon, which is $\$5.30/(1.05113)^2 = \4.797 .
- The cash flows in the 3-year bond occurring in year 3 must have a present value equal to $\$100 - \$5.048 - \$4.797 = \90.155 .
- The 3-year zero coupon rate must satisfy

$$\$90.155 = \frac{\$105.30}{[1 + zc(3)]^3},$$

or $zc(3) = 5.312\%$.

- To find the 4-Year Zero Coupon Rate:

- Working now with the 4-year bond that has a coupon of 5.47%, use the 1-year zero coupon rate to find the present value of the first-year coupon, which is $\$5.47/1.05 = \5.210 ; use the 2-year zero coupon rate to find the present value of the second-year coupon, which is $\$5.47/(1.05113)^2 = \4.951 ; and use the 3-year zero coupon rate to find the present value of the third-year coupon, which is $\$5.47/(1.05312)^3 = \4.683 .
- The cash flows in the 4-year bond occurring in year 4 must have a present value equal to $\$100 - \$5.210 - \$4.951 - \$4.683 = \$85.156$.
- The 4-year zero coupon rate must satisfy

$$\$85.156 = \frac{\$105.47}{[1 + zc(4)]^4},$$

or $zc(4) = 5.494\%$.

From Zero Coupon Rates to Forward Rates

Once we have found the zero coupon rates, finding forward rates is a piece of cake. Let $f(t, T)$ denote the $T - t$ year forward rate from time t to time T . Then the forward rate $f(t, T)$ must satisfy

$$[1 + zc(t)]^t \times [1 + f(t, T)]^{T-t} = [1 + zc(T)]^T. \quad (3.6)$$

For our purposes here we will only need to solve for 1-year forward rates, and thus we will make use of Equation 3.6 in the following form:

$$[1 + zc(t)]^t \times [1 + f(t, t+1)]^1 = [1 + zc(t+1)]^{t+1}. \quad (3.7)$$

Maturity	Swap Rate	1-Year Bond	2-Year Bond	3-Year Bond	4-Year Bond
1	5.00%	100.000	4.867	5.048	5.210
2	5.11%		95.133	4.797	4.951
3	5.30%			90.155	4.683
4	5.47%				85.156
Zero Coupon Rates		5.000%	5.113%	5.312%	5.494%
Forward Rates		5.000%	5.226%	5.712%	6.041%

Table 3.7: Zero Coupon Rates and Forwards Obtained from Bootstrapping

For example, since the 1-year zero coupon rate is 5% and the 2-year zero coupon rate is 5.113%, the 1-year forward in one year, $f(1, 2)$, must satisfy

$$(1.05)^1 \times [1 + f(1, 2)]^1 = (1.05113)^2,$$

or $f(1, 2) = 5.226\%$. The other 1-year forwards are found similarly making use of Equation 3.7. The results of this bootstrapping technique are shown in Table 3.7. Although we just worked through an example that was four years long, this technique can be used to find zero coupon and forward rates from swap curves with longer maturities as well. By the way, how did we come up with $f(0, 1)$, that first 1-year forward, the 1-year forward starting today, as being 5%?¹⁶

So how do we interpret each of these 1-year forwards that we just calculated? These are forward rates of interest that one could lock in and they are implicit in the swap curve. As is customary in the industry, we will regard these forwards as the expectation of where 1-year LIBOR rates will set each year in the future. For example, since $f(2, 3) = 5.712\%$, we will say that the market expects 1-year LIBOR to be 5.712% two years from now.

Figure 3.4 shows graphically the relationship between the zero coupon rates and the forward rates that we derived from the swap curve in Table 3.5. As shown in this figure, the 1-year forward rates increase in each successive year. We should not allow this particular example to lull us into the belief that the forward rates are always a monotonically increasing function of time when the swap curve is upward sloping. Consider, for example, the swap curve in Table 3.8. One thing to note is that this curve does not contain a

¹⁶A forward rate out of today is just a spot rate, which in this case is today's 1-year LIBOR rate of 5%.

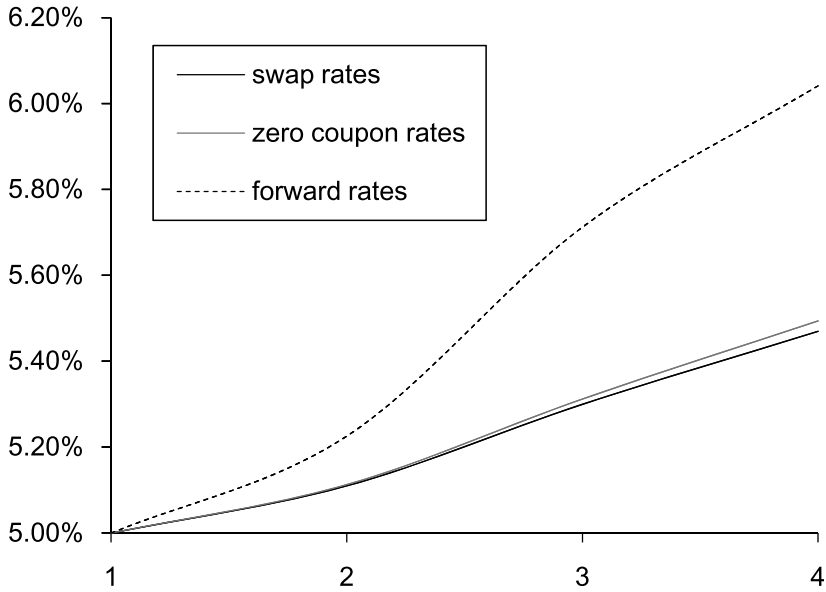


Figure 3.4: Using Swap Rates to Get Zeros and Forwards

Maturity	Swap Rate
Year 1	2.00%
Year 2	2.10%
Year 3	2.20%
Year 4	2.40%
Year 5	2.60%
Year 10	2.75%
Year 30	3.00%

Table 3.8: Sample Swap Curve

par swap rate for each year. Let's assume for the moment that we use linear interpolation of the nearby, surrounding par swap rates to generate swap rates that do not appear explicitly in this curve. Thus, for example, the 6-year swap rate is assumed to be 2.63% (i.e., an 80% weighting of the 5-year swap rate and a 20% weighting of the 10-year swap rate), the 7-year swap rate is

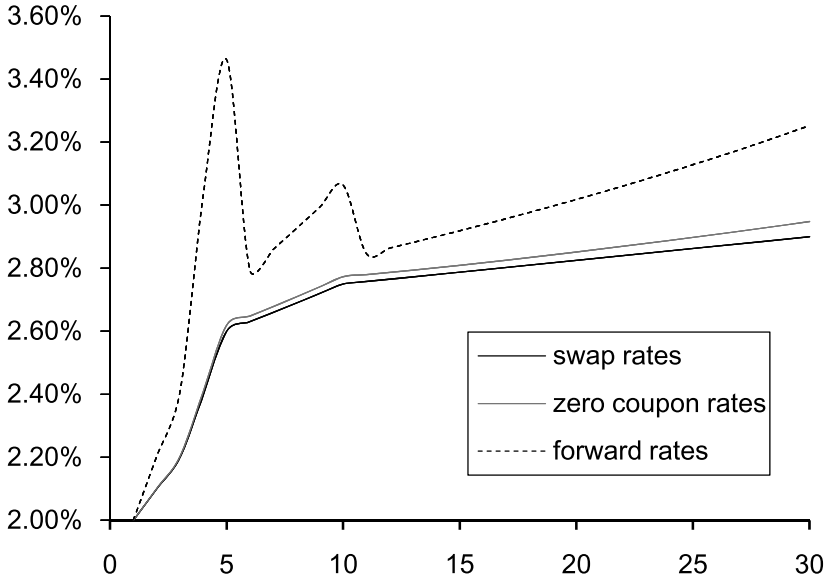


Figure 3.5: Forward Rates Obtained from Bootstrapping the Swap Curve in Table 3.8

assumed to be 2.66%, etc.¹⁷ Using the same bootstrapping technique we have just developed, the forward rates for this swap curve are shown in Figure 3.5.

The resultant curve of 1-year forwards is not “pleasing.” These forwards appear “well behaved” for the first five years, but the jagged profile that follows with significant oscillation thereafter is deemed undesirable by many in the industry. More sophisticated techniques have been adapted throughout the industry to create a smoother pattern of forward LIBOR rates, which is generally considered more desirable.¹⁸ Yet even with more advanced methods, given an upward-sloping swap curve such as the one in Table 3.8, the

¹⁷Suppose s_L and s_H are swap rates of maturity L and H , respectively, and we wish to compute an M -year swap rate, s_M , where $L < M < H$, by linear interpolation. Then s_M is given by

$$s_M = s_L + \left(\frac{M - L}{H - L} \right) \times (s_H - s_L).$$

¹⁸The issue of appropriate and adequate swap curve construction is a big one of the industry. Many different methods are employed. See, for example, Hagan (2003), Ron (2000), and Zhou (2002). While generic methodologies abound, many institutions have spent considerable time developing proprietary methods and systems for curve construction.

usual pattern of forward LIBOR rates that are obtained start out increasing but eventually decrease at some point in the future.

Creating a Swap Model

We are going to use this bootstrapping methodology to create a swap model. This methodology is very important and will serve as the basis of pricing most of the trades we will look at in this and subsequent chapters. It is important to note at the outset that the swap model we will develop here, while a powerful tool, will not be a fully robust model. Any comprehensive swap model would have to have the capability of modeling swap rates (and zero coupon rates and forward rates) to any particular date. Our model will only have the capacity to consider annual cash flows. A more realistic model that allowed for the flexibility of incorporating swap cash flows on any arbitrarily specified date would use a similar bootstrapping methodology as we have developed, albeit with more sophisticated techniques for interpolating rates beyond a set of benchmark par swap rates such as given in Table 3.8. Despite these shortcomings, our model will allow us to price a great variety of trades common in the swaps market.

Let's return now to our stylized example and consider the swap curve given in Table 3.5. We will use the zero coupon and forward rates derived from bootstrapping to price up a \$100mm 4-year swap in which we are receiving fixed. To price up a swap in this context means to figure out what the fixed coupon in the trade should be. Remember, the swap rate is the fixed rate that makes the present value of the fixed side equal to the present value of the floating side. We use the zero coupon rates to compute present values, and we use the forwards to determine the market's expectation of 1-year LIBOR over time.

Figure 3.6 shows a swap model used to price up the 4-year swap. Each row corresponds to a given year in the swap, and contains the appropriate zero coupon rate that we use to discount cash flows that occur at the end of that year and also contains the forward rate applicable for that year. For example, let's look at the last row in the model that refers to cash flow received in year 4. The 4-year zero coupon rate, $zc(4)$, of 5.494% is provided and is used to discount all cash flows received in year 4. This row also contains $f(3, 4) = 6.041\%$, the 1-year forward LIBOR in the fourth year. We can think of this as the market's expectation of where 1-year LIBOR will reset at year 3, for accrual and payment at year 4. Based on this forward rate, we see that the floating cash flow that we expect in the final year of the swap is $\$100\text{mm} \times f(3, 4) = \$6,041,364$. The present value of this expected floating flow is

$$\frac{\$100\text{mm} \times f(3, 4)}{[1 + zc(4)]^4} = \$4,877,792.$$

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	5,470,000	5,209,524	5,000,000	4,761,905
2	5.113%	5.226%	5,470,000	4,950,807	5,225,748	4,729,739
3	5.312%	5.712%	5,470,000	4,683,288	5,712,202	4,890,656
4	5.494%	6.041%	5,470,000	4,416,473	6,041,364	4,877,792
				19,260,092		19,260,092
		Fixed rate	5.470%	NPV	0	
		Notional	100,000,000			

Figure 3.6: A Swap Model

The other nominal floating cash flows are modeled, and the present values of these cash flows are calculated in a similar manner.¹⁹ The sum of the present values of each of the floating coupons is \$19,260,092. Our job, then, is to iterate and solve for that single fixed coupon such that the sum of the present value of the fixed cash flows is also equal to \$19,260,092. In Figure 3.6, we have solved for the fixed coupon that makes the present value of the fixed side equal to the present value of the floating side. That fixed coupon is 5.47%.

With a fixed coupon of 5.47%, the nominal cash flows on the fixed side of the swap are $\$100\text{mm} \times 5.47\% = \$5,470,000$ each year. The zero coupon rates are used to compute the present value of these nominal fixed flows. For example, the present value of the fixed coupon received at the end of year 3 is

$$\frac{\$100\text{mm} \times 5.47\%}{[1 + zc(3)]^3} = \$4,683,288.$$

Adding up the present values of each of the fixed coupons of 5.47% gives us \$19,260,092, the same present value as the floating side of the swap. Thus, the 4-year swap rate is indeed 5.47%.

¹⁹The first floating cash flow is calculated off of today's spot LIBOR setting. As we argued earlier, in our stylized example with annual bond coupons on both sides of the swap, the spot 1-year LIBOR rate is equal to the 1-year swap rate. This happens to be 5% in this numerical example. And as discussed before, the spot 1-year LIBOR is the (degenerate) 1-year forward that sets today, $f(0,1)$.

So what does this tell us? It's like that *Seinfeld* episode: nothing, absolutely nothing.²⁰ We just found using our swap model that the 4-year swap rate is 5.47%. But recall that when we started this whole thing, we were given a swap curve in Table 3.5 in which it was given that the 4-year swap rate is 5.47%. So we went through a lot of work just to show that something was exactly what we knew it to be. This would seem, then, to be a big waste of time. But although the answer to this particular problem was readily available before we started modeling, the technology that we have just developed will be crucial for pricing a variety of swaps, as we will soon see. So perhaps it's not that big a waste of time after all.

Relationship Between Swap Rate and Forwards

Figure 3.7 shows the relationship between the 4-year swap rate that we just solved for and the forwards. As this figure demonstrates, the *forward curve* (the relationship between successive 1-year LIBOR forwards and time) is upward sloping, and the swap rate that we solved for lies somewhere in the middle of the forwards. It might be tempting to say that the swap rate is a simple average of the forwards, but this is not the case. However, it is accurate to say that the swap rate that we computed is a specific weighted average of the forwards. In fact, define

$$DF(0, t) = \frac{1}{[1 + zc(t)]^t},$$

the *discount factor for time t* , and further define

$$A = \sum_{t=1}^4 DF(0, t),$$

the sum of the discount factors. This quantity A is the annuity factor; it is the number that, when multiplied by \$1, gives then present value of a \$1 annuity received over the life of the swap. Now define weights w_t , for $t = 1, 2, 3$, and 4 by

$$w_t = \frac{DF(0, t)}{A}.$$

It is easy to show that these weights sum to one. Note that w_t is the proportion of the present value of the annuity of \$1 that is associated with a

²⁰Reference to Episodes 63 and 64, "The Pilot (1)" and "The Pilot (2)," respectively, in the sitcom *Seinfeld*. In these episodes, the characters George Costanza (played by Jason Alexander) and Jerry Seinfeld (played by himself) get permission from NBC to go ahead with their idea for a new sitcom that is about nothing.